Problems on Multiquadrant Operation

Problem (2.2). A Motor is used to drive a hoist. Motor characteristics are given by:

Quadrants i, ii and IV is T = 200 – 0.2N N.M.

Quadrants ii, iii and IV is T = -200 – 0.2N N.M.

Where N is the speed in rpm

When hoist is loaded, the net load torque T_L = 100 n.m, and when is unloaded, net load torque T_L= -80 n-m. Obtain the equilibrium speeds for operation in all the four quadrants.

Sol: Given data,

Motor torque characteristics in quadrants i, ii and IV is

T_1= 200 – 0.2N N-M. ……. (1)

Motor torque characteristics in quadrants ii, iii and IV are

T_2 =-200 – 0.2N N-M. ……. (2)

Net load torque when hoist is loaded T_{L1} = 100 N.M.

Net load torque when hoist is unloaded T_{L2} = -80 N.M.

Speeds in all quadrants N1, N2, N3, N4 =??

The boundary conditions for equation (1) are

If T=0 (i.e at no-load)→ 0 = 200 – 0.2N

→ 0.2N = 200 → N = 1000 rpm

ω_m = 104.719 rad/sec.

If N=0 ; i.e at zero speed

(1) T=200-0.2(0) = 200 N.M

i.e equation (1) represents a straight line passing through the points (0,1000) and (200,0) respectively.

The boundary condition for equation (2) are ; I,e

T=0, i.e at no-load

(2) 0= -200-0.2N

0.2N = -200

N = -1000 rpm → ω_m = -104.719 rad/sec

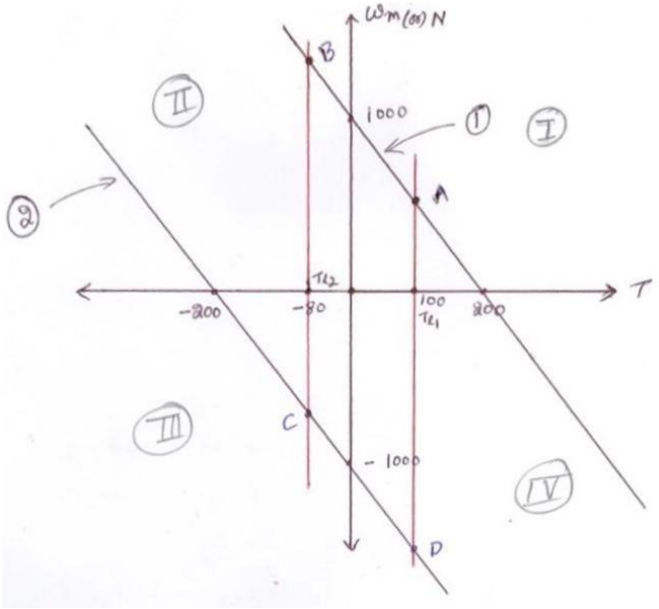
N=0 ; i.e at zero speed

(3) T = =200 – 0.2(0)

T = -200 N.M

i.e equation (2) represents a straight line passing through the points (0,-1000) and (-200,0) respectively.

From boundary conditions the equations (1) & (2) and load torques can be drawn as shown below.



From the above diagram A,B,C,D are the equilibrium points in quadrant I,II,III,IV respectively

At equilibrium point “A” the fundamental torque equation is

$T_{L1} = T_1$

$\rightarrow 100 = 200 - 0.2\ N_A \rightarrow 0.2N = 100 \rightarrow \quad N_A = 500$ rpm

At the equilibrium “point B” fundamental torque equation is:

$T_{L2} = T_1$

$-80 = 200 – 0.2N_B$

$\rightarrow 0.2N = 280$

$N_B = 280/0.2 = 1400$

$N_B = 1400$ rpm

At the equilibrium “point C” fundamental torque equation is:

$T_{L2} = T_2$

$-80 = -200 – 0.2Nc$

$0.2N_C = -120$

$N_C = -120/0.2 = -600$ rpm

At the equilibrium “point D” fundamental torque equation is:

$T_{L1} = T_2$

$100 = -200 – 0.2ND$

$0.2N_D = -300$

$N_D = -300/0.2 = -1500$ rpm

For the given hoist the equilibrium speed for operation in all four quadrants are:

i. Quadrant N_A = 500 RPM
ii. Quadrant N_B = 1400 RPM
iii. Quadrant N_c = -600 RPM
iv. Quadrant N_D = -1500 RPM

Problem (2.10). A motor-load system has the following details: Quadrants i and ii, T=400 – 0.4N, N.M where N is the speed in rpm. Motor is coupled to an active load torque T_L= ± 200 N-M. Calculate the motor speeds for motoring and braking operations in the forward direction. When operating in quadrants iii & iv, T = -400 – 0.4N, N-M. Calculate the equilibrium speed in Quadrant iii.

Sol: Given data,

Motor torque equation in quadrants i & ii is:

$$T_1 = 400 - 0.4N_1 \text{ .....(1)}$$

Motor torque equation in quadrant iii & iv is:

$$T_2 = -400 - 0.4N_2$$

Active load torque TL = ± 200 ,N.M.

"Active load means load torque retains sign when the direction of the drive rotation is changed"

Motor speed in forward motoring operation N_A =?

Motor speed in forward braking operation N_B =?

Motor speed in iii quadrant N_c =?

For the equation (1) the boundary conditions are:

At T1 = 0 i,e at No-Load

(1) → $0 = 400 - 0.4N_1$

→ 0.4N1 = 400 → N1 = 1000 rpm

At N_1 = 0 i,e at zero speed

$T_1 = 400 - 0.4(0)$

T_1 = 400 N.M

Equation (1) represents a straight line passing through (0, 1000) & (400,0)

For equation (2) the boundary conditions are:

At T_2 = 0 , i.e at No-Load

(1) → $0 = -400 - 0.4N_2$

→ $0.4N_2 = -400$

N_2 = -1000rpm

At N_2 = 0 i,e at zero speed

(2) → $T_2 = -400 - 0.4(0)$ → T_2 = -400 N.M

Equation (2) represents the line passing through (0,-1000) & (-400, 0)

The representation of lines (1) & (2) are as shown below:

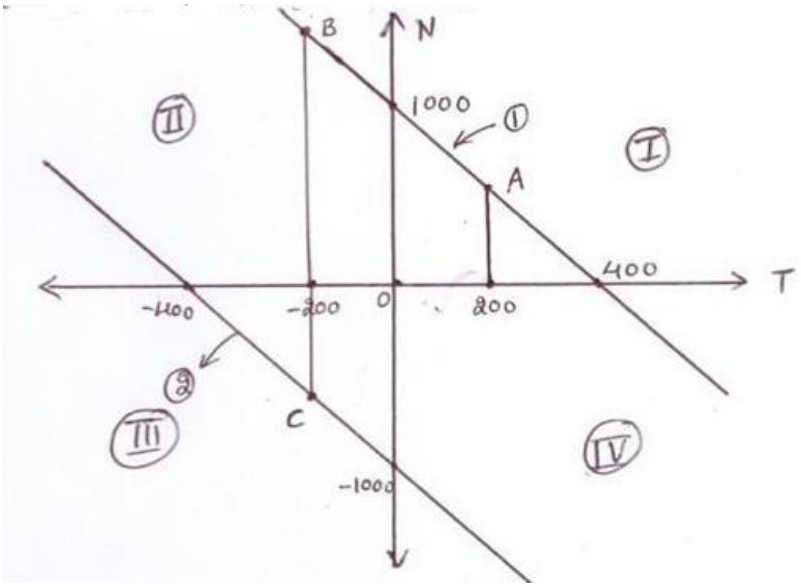
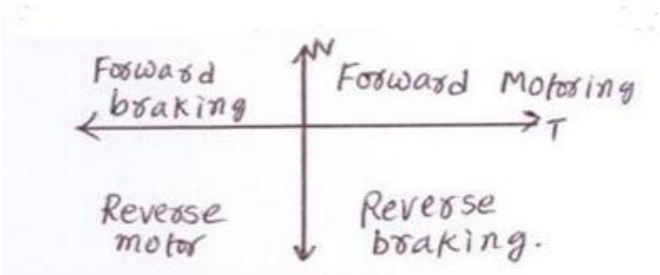


Figure 1 Representation of lines 1,2

The four-quadrant operation of an active load is shown below:



→ Point "A" represents operating point for forward motoring operation.

→ Point "B" represents operating point for forward braking operation.

→ Point "C" represents operating point in quadrant "3" since it is an active load hence it remains –ve even if the direction of rotation is changed.

Speed at operating point "A" is "N_A" is get from eq (1)

$200 = 400 – 0.4N_A$

$0.4N_A = 200$

$N_A = 500$ rpm

Speed at operating point "B" is N_B from eq (1)

$-200 = 400 – 0.4N_B$

$0.4N_B = 600$

$N_B = 600/0.4 = 1500$ rpm

Speed at operating point "C" is Nc from eq (2)

$-200 = -400 – 0.4N_c$

$0.4Nc = -200 \rightarrow N_c = 600/0.4 = 1500$ rpm

For the system:

- **Speed in forward motor operation N_A = 500 rpm.**
- **Speed in forward braking operation N_B = 1500 rpm.**
- **Speed in iii quadrant is N_c = -500 rpm.**